Info

- no tuts ts week

- no tuts switching

Classes of materials

	Polymers	Ceramics	Metals
ductility	Varies	Poor	Good
Conductivity	low	low	...
hardness/strength	low-medium	very high	fair
corrosion res	fair-good	...	fair-poor

Bonding

- Properties of materials (strength, hardness, conductivity) are determined by bonding

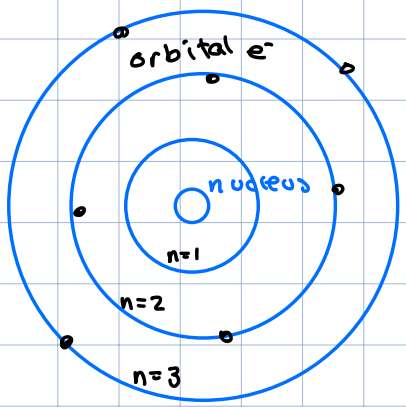
- Bonding is determined by:

  - e^- structure (distribution of e^- in atomic orbitals)

  - # of e^- and electronegativity (tendency for

or atom to attract or e^-

atomic structure



- Bohr model (1913)

- Z = # of p^+

- N = # of n^0

- Atomic mass $A \approx Z + N$

- Isotopes: same Z, diff N → same element
  diff atomic mass (^{12}C & ^{13}C)

- Reported atomic mass = abundance-weighted avg of the isotope

ex: C: $0.989(12.000) + 0.011(13.003) = 12.011 \frac{g}{mol}$

Periodic table

^{13} Al → element
26.982 → atomic mass

^{13} → Z

Electronegativity

- ↑ en = strong tendency to accept e^- (F, Cl)
- ↓ en = strong tendency to give up e^- (Li, Na, K)

terms

- Density (g/cm^3) ~ most solids range 1-23 g/cm^3 (ρ)
- Atomic # = Z
- Atomic weight = A (g/mol)
- Mole = # of particle → $N_A = 6.022 \times 10^{23}$ mol/part

Review problem

how many atoms in 6 gram of carbon

$\rho = 2.26\ g/cm^3$ $\quad 1\ mol_C = 12.011\ g/mol$

$$n = \frac{m}{M} \rightarrow n = \frac{6g}{12.011 g/mol} \approx 0.5\ mol$$

$$n \cdot N_A = \#\ atoms \rightarrow 0.5\ mol \cdot N_A = 3.011 \times 10^{23}\ atoms$$

Calculate the Volume of 1 mol of Au

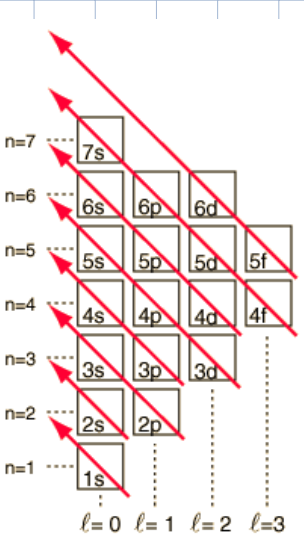
$\rho = 19.32\ g/cm^3 \qquad n = 1\ mol$

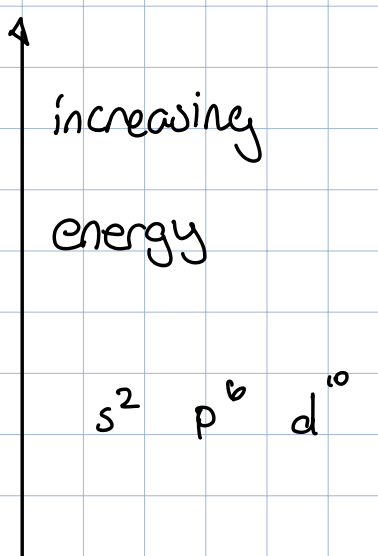
$n \cdot M = m \rightarrow 1\ mol \cdot 196.6654\ g/mol = m$

$196.665\ g = m$

$\frac{m}{\rho} = V \rightarrow \frac{196.665\ g}{19.32\ g/cm^3} = V = 10.194\ cm^3$

Energy states

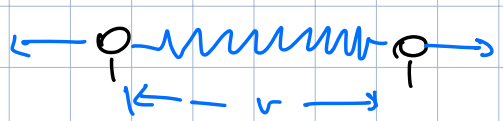




Stable e^- config

- complet s & p shells
- tend to be unreactive

Bonding Forces & Energies



r = bond length

- net force is given by the sum of an attractive and a repulsive force

- Potential E is given by the integral of the net force curve w/ respect to distance

$$E = \int F \cdot dr$$

note equilibrium seperation occurs where the net force = 0

- Bonding E : Min of the potential vs distance curve

  - Indicates how much E must be supplied to completely dissociate the 2 atoms

  - Depth well

    → Deep well = strongly bonded

    → Shallow well = weakly bonded

Bond E

- the ↑ the bond E
  - the ↑ the melting temp
  - the ↑ the hardness
- state as function of bonding E
  - solid (↑)
  - liquid (intermediate)
  - gas (↓ low)